

ELY, MN, USA

WMO: 727459

Lat: 47.817N

Lon: 91.833W

Elev: 1456

StdP: 13.94

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94964

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.5	-19.9	-34.6	0.8	-26.8	-27.1	1.3	-18.9	19.0	11.6	17.3	12.1	1.3	250	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	84.2	68.8	81.6	67.0	79.2	64.9	73.2	80.6	70.3	78.2	68.0	75.4	9.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.5	118.4	77.8	67.7	107.3	74.7	65.5	99.4	72.9	37.8	80.2	35.1	77.3	33.2	75.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.0	16.0	13.5	-33.5	89.9	6.8	3.5	-38.4	92.5	-42.4	94.5	-46.3	96.5	-51.2	99.1		
			-33.5	76.8	6.3	5.5	-38.0	80.8	-41.6	84.0	-45.1	87.1	-49.7	91.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.2	6.6	10.9	23.9	37.5	50.6	60.5	65.3	63.0	55.4	42.4	27.7	13.5
	DBStd	22.94	14.79	13.94	13.35	9.69	8.70	6.89	6.06	6.14	8.42	8.73	11.50	13.54
	HDD50	5857	1346	1094	812	390	97	5	0	1	39	271	669	1133
	HDD65	9956	1811	1514	1275	826	455	166	73	113	306	701	1118	1598
	CDD50	1566	0	0	3	14	116	320	473	403	200	36	1	0
	CDD65	191	0	0	0	0	9	32	81	50	18	1	0	0
	CDH74	2085	0	0	0	6	163	406	812	523	162	13	0	0
	CDH80	455	0	0	0	0	36	95	188	106	29	1	0	0
Wind	WSAvg	5.4	5.2	5.2	5.5	5.8	6.1	5.3	4.9	4.7	5.4	6.0	5.8	5.0
Precipitation	PrecAvg	29.0	0.9	0.7	1.1	2.2	3.0	4.0	4.0	3.2	4.0	2.9	1.6	1.1
	PrecMax	36.0	2.3	1.8	2.3	5.0	4.7	7.8	7.8	7.1	10.8	5.3	4.0	2.1
	PrecMin	23.2	0.3	0.2	0.4	0.6	1.2	0.7	1.7	0.9	0.8	1.2	0.3	0.2
	PrecStd	3.5	0.5	0.4	0.6	1.3	0.9	1.7	1.6	1.5	2.4	1.1	0.9	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.2	46.2	63.0	72.7	84.2	89.3	89.6	88.0	82.9	75.0	59.3	43.2
		MCWB	33.6	39.7	52.3	55.0	63.8	74.5	71.8	71.9	68.0	60.8	50.5	36.9
	2%	DB	34.4	40.7	54.0	65.7	79.1	82.0	84.2	82.2	79.1	67.8	53.9	37.4
		MCWB	32.3	36.2	44.7	50.5	59.7	65.9	70.3	68.1	65.6	56.7	48.4	34.4
	5%	DB	30.4	36.1	47.7	61.1	73.0	79.0	81.5	79.5	74.6	63.0	46.5	35.6
		MCWB	29.2	32.5	40.9	46.7	56.9	63.4	68.0	66.1	63.7	54.2	41.5	33.4
	10%	DB	27.6	31.8	42.9	55.1	68.2	75.0	79.0	76.6	71.1	56.8	43.4	32.1
		MCWB	26.0	29.7	36.3	43.2	54.8	61.7	65.9	65.0	61.7	49.5	38.8	30.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.9	41.3	54.8	58.6	68.1	75.4	76.5	75.7	72.8	63.4	53.6	38.3
		MCDB	38.7	45.1	61.2	68.1	77.3	83.9	83.9	83.2	80.0	69.7	57.4	41.0
	2%	WB	32.4	36.6	46.1	51.5	63.2	70.2	73.6	71.3	68.5	58.8	47.5	35.5
		MCDB	34.3	40.3	51.3	62.5	73.2	79.1	80.6	78.9	75.1	65.6	52.7	38.0
	5%	WB	29.3	32.7	40.9	48.0	59.5	67.2	70.8	68.7	65.8	54.7	42.8	32.6
		MCDB	30.5	35.6	47.4	59.3	69.7	74.4	78.3	76.5	71.7	62.2	46.7	34.6
	10%	WB	26.1	29.0	36.8	44.4	56.5	64.2	68.2	66.8	63.1	50.8	39.2	30.3
		MCDB	27.6	31.1	41.9	54.4	65.4	71.0	75.7	74.0	68.8	56.1	43.1	31.6
Mean Daily Temperature Range	5% DB	MDBR	20.4	22.6	22.5	22.3	23.8	22.4	22.1	22.3	20.9	16.9	14.5	16.3
		MCDBR	18.3	22.3	25.4	32.2	32.3	27.8	24.6	25.9	26.1	26.1	20.4	18.2
		MCWBR	16.1	17.6	16.9	19.1	16.8	14.2	12.8	13.9	15.2	16.1	14.6	15.0
	5% WB	MCDBR	17.1	18.8	24.0	28.9	26.1	22.0	21.8	22.1	20.6	23.6	19.2	16.1
	MCWBR	15.0	14.9	17.0	18.3	14.8	13.2	13.1	13.5	13.5	15.8	14.4	13.7	
Clear-Sky Solar Irradiance	taub	0.249	0.264	0.307	0.335	0.355	0.370	0.383	0.369	0.352	0.308	0.284	0.257	
	taud	2.381	2.345	2.375	2.423	2.418	2.394	2.383	2.451	2.474	2.582	2.528	2.428	
	Ebn at Noon	270	289	291	292	288	283	277	277	272	271	255	252	
	Edh at Noon	21	28	32	34	35	36	36	33	29	22	19	18	
All-Sky Solar Radiation	RadAvg	407	745	1138	1489	1691	1807	1904	1614	1188	707	391	302	
	RadStd	51	79	105	150	115	135	89	115	84	88	49	30	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-95
(47)	Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page